

Chapter 17

Differentiation of Trigonometric Functions

RADIAN MEASURE. Let s denote the length of an arc AB intercepted by the central angle AOB on a circle of radius r , and let S denote the area of the sector AOB (see Fig. 17-1). (If s is $\frac{1}{360}$ of the circumference, then angle AOB has measure 1° ; if $s = r$, angle AOB has measure 1 radian (rad). Recall that $1 \text{ rad} = 180/\pi$ degrees and $1^\circ = \pi/180$ rad. Thus, $0^\circ = 0 \text{ rad}$; $30^\circ = \pi/6 \text{ rad}$; $45^\circ = \pi/4 \text{ rad}$; $180^\circ = \pi \text{ rad}$; and $360^\circ = 2\pi \text{ rad}$.)

Suppose $\angle AOB$ is measured as α degrees; then

$$s = \frac{\pi}{180} \alpha r \quad \text{and} \quad S = \frac{\pi}{360} \alpha r^2 \quad (17.1)$$

Suppose next that $\angle AOB$ is measured as θ radians; then

$$s = \theta r \quad \text{and} \quad S = \frac{1}{2} \theta r^2 \quad (17.2)$$

A comparison of (17.1) and (17.2) will make clear one of the advantages of radian measure.

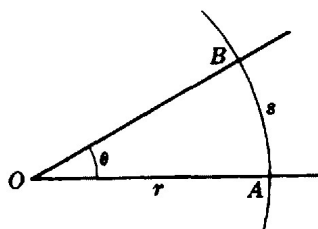


Fig. 17-1

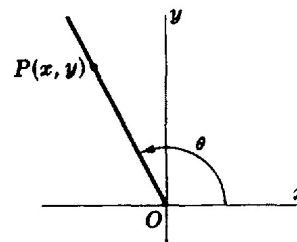


Fig. 17-2

TRIGONOMETRIC FUNCTIONS. Let θ be any real number. Construct the angle whose measure is θ radians with vertex at the origin of a rectangular coordinate system, and initial side along the positive x axis (Fig. 17-2). Take $P(x, y)$ on the terminal side of the angle a unit distance from O ; then $\sin \theta = y$ and $\cos \theta = x$. The domain of definition of both $\sin \theta$ and $\cos \theta$ is the set of real numbers; the range of $\sin \theta$ is $-1 \leq y \leq 1$, and the range of $\cos \theta$ is $-1 \leq x \leq 1$. From

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \quad \text{and} \quad \sec \theta = \frac{1}{\cos \theta}$$

For both $\tan \theta$ and $\sec \theta$ the domain of definition ($\cos \theta \neq 0$) is $\theta \neq \pm \frac{2n-1}{2} \pi$, ($n = 1, 2, 3, \dots$). It is left as an exercise for the reader to consider the functions

$$\cot \theta = \frac{\cos \theta}{\sin \theta} \quad \text{and} \quad \csc \theta = \frac{1}{\sin \theta}$$

Recall that, if θ is an acute angle of a right triangle ABC (Fig. 17-3), then

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}} = \frac{BC}{AB} \quad \cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}} = \frac{AC}{AB} \quad \tan \theta = \frac{\text{opposite side}}{\text{adjacent side}} = \frac{BC}{AC}$$

The slope m of a nonvertical line is equal to $\tan \alpha$, where α is the counterclockwise angle from the positive x axis to the line. (See Fig. 17-4.)

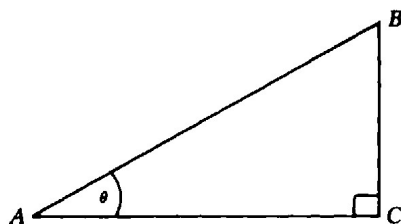


Fig. 17-3

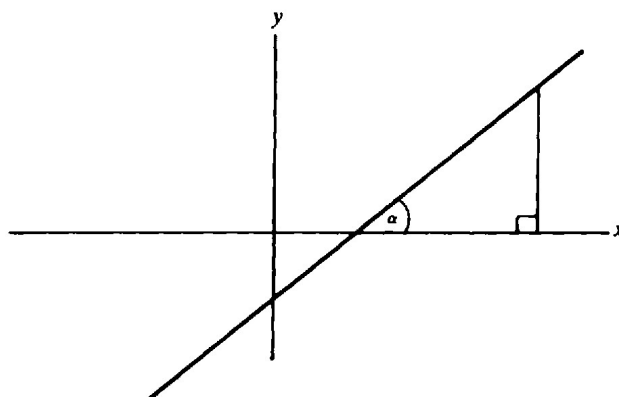


Fig. 17-4

Table 17-1 lists some standard trigonometric identities, and Table 17-2 contains some useful values of the trigonometric functions.

Table 17-1

$\sin^2 \theta + \cos^2 \theta = 1$
$\sin(-\theta) = -\sin \theta, \cos(-\theta) = \cos \theta$
$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$
$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$
$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$
$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$
$\sin 2\alpha = 2 \sin \alpha \cos \alpha$
$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = 1 - 2 \sin^2 \alpha = 2 \cos^2 \alpha - 1$
$\sin(\alpha + 2\pi) = \sin \alpha, \cos(\alpha + 2\pi) = \cos \alpha$
$\sin(\alpha + \pi) = -\sin \alpha, \cos(\alpha + \pi) = -\cos \alpha, \tan(\alpha + \pi) = \tan \alpha$
$\sin\left(\frac{\pi}{2} - \alpha\right) = \cos \alpha, \cos\left(\frac{\pi}{2} - \alpha\right) = \sin \alpha$
$\sin(\pi - \alpha) = \sin \alpha, \cos(\pi - \alpha) = -\cos \alpha$
$\sec^2 \alpha = 1 + \tan^2 \alpha$
$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$
$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$

Table 17-2

x	$\sin x$	$\cos x$	$\tan x$
0	0	1	0
$\pi/6$	$1/2$	$\sqrt{3}/2$	$\sqrt{3}/3$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1
$\pi/3$	$\sqrt{3}/2$	$1/2$	$\sqrt{3}$
$\pi/2$	1	0	∞
π	0	-1	0
$3\pi/2$	-1	0	∞

In Problem 1, we prove that

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

(Had the angle been measured in degrees, the limit would have been $\pi/180$. This is another reason why radian measure is always used in the calculus.)

DIFFERENTIATION FORMULAS

$$14. \quad \frac{d}{dx} (\sin x) = \cos x$$

$$15. \quad \frac{d}{dx} (\cos x) = -\sin x$$

$$16. \quad \frac{d}{dx} (\tan x) = \sec^2 x$$

$$17. \quad \frac{d}{dx} (\cot x) = -\csc^2 x$$

$$18. \quad \frac{d}{dx} (\sec x) = \sec x \tan x$$

$$19. \quad \frac{d}{dx} (\csc x) = -\csc x \cot x$$

(See Problems 2 to 23.)

Solved Problems

1. Prove: (a) $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$ and (b) $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0$.

(a) Since $\frac{\sin(-\theta)}{-\theta} = \frac{\sin \theta}{\theta}$, we need consider only $\lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta}$. In Fig. 17-5, let $\theta = \angle AOB$ be a small positive central angle of a circle of radius $OA = 1$. Denote by C the foot of the perpendicular dropped from B onto OA , and by D the intersection of OB and an arc of radius OC . Now

$$\text{Sector } COD \leq \triangle COB \leq \text{sector } AOB$$

so that

$$\frac{1}{2} \theta \cos^2 \theta \leq \frac{1}{2} \sin \theta \cos \theta \leq \frac{1}{2} \theta$$

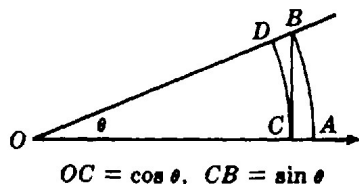


Fig. 17-5

Dividing by $\frac{1}{2}\theta \cos \theta > 0$, we obtain

$$\cos \theta \leq \frac{\sin \theta}{\theta} \leq \frac{1}{\cos \theta}$$

Let $\theta \rightarrow 0^+$; then $\cos \theta \rightarrow 1$, $\frac{1}{\cos \theta} \rightarrow 1$, and $1 \leq \lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta} \leq 1$; hence, $\lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta} = 1$.

$$\begin{aligned} (b) \quad \lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} &= \lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} \frac{\cos \theta + 1}{\cos \theta + 1} \\ &= \lim_{\theta \rightarrow 0} \frac{\cos^2 \theta - 1}{\theta(\cos \theta + 1)} = \lim_{\theta \rightarrow 0} -\frac{\sin^2 \theta}{\theta(\cos \theta + 1)} \\ &= -\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\cos \theta + 1} = -(1)\left(\frac{0}{2}\right) = 0 \end{aligned}$$

2. Derive: $\frac{d}{dx} (\sin x) = \cos x$.

$$\begin{aligned} \text{Let } y &= \sin x. \text{ Then } y + \Delta y = \sin(x + \Delta x) \text{ and} \\ \Delta y &= \sin(x + \Delta x) - \sin x = \cos x \sin \Delta x + \sin x \cos \Delta x - \sin x \\ &= \cos x \sin \Delta x + \sin x(\cos \Delta x - 1) \\ \frac{dy}{dx} &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \left(\cos x \frac{\sin \Delta x}{\Delta x} + \sin x \frac{\cos \Delta x - 1}{\Delta x} \right) \\ &= (\cos x) \lim_{\Delta x \rightarrow 0} \frac{\sin \Delta x}{\Delta x} + (\sin x) \lim_{\Delta x \rightarrow 0} \frac{\cos \Delta x - 1}{\Delta x} \\ &= (\cos x)(1) + (\sin x)(0) = \cos x \end{aligned}$$

3. Derive: $\frac{d}{dx} (\cos x) = -\sin x$.

$$\frac{d}{dx} (\cos x) = \frac{d}{dx} \left[\sin \left(\frac{\pi}{2} - x \right) \right] = -\cos \left(\frac{\pi}{2} - x \right) = -\sin x$$

4. Derive: $\frac{d}{dx} (\tan x) = \sec^2 x$.

$$\begin{aligned} \frac{d}{dx} (\tan x) &= \frac{d}{dx} \left(\frac{\sin x}{\cos x} \right) = \frac{\cos x \cos x - \sin x (-\sin x)}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x \end{aligned}$$

In Problems 5 to 12, find the first derivative.

$$5. \quad y = \sin 3x + \cos 2x: \quad y' = \cos 3x \frac{d}{dx} (3x) - \sin 2x \frac{d}{dx} (2x) = 3 \cos 3x - 2 \sin 2x$$

$$6. \quad y = \tan x^2: \quad y' = \sec^2 x^2 \frac{d}{dx} (x^2) = 2x \sec^2 x^2$$

$$7. \quad y = \tan^2 x = (\tan x)^2: \quad y' = 2 \tan x \frac{d}{dx} (\tan x) = 2 \tan x \sec^2 x$$

$$8. \quad y = \cot (1 - 2x^2): \quad y' = -\csc^2 (1 - 2x^2) \frac{d}{dx} (1 - 2x^2) = 4x \csc^2 (1 - 2x^2)$$

$$9. \quad y = \sec^3 \sqrt{x} = \sec^3 x^{1/2}:$$

$$y' = 3 \sec^2 x^{1/2} \frac{d}{dx} (\sec x^{1/2}) = 3 \sec^2 x^{1/2} \sec x^{1/2} \tan x^{1/2} \frac{d}{dx} (x^{1/2}) = \frac{3}{2\sqrt{x}} \sec^3 \sqrt{x} \tan \sqrt{x}$$

10. $\rho = \sqrt{\csc 2\theta} = (\csc 2\theta)^{1/2}$:

$$\rho' = \frac{1}{2} (\csc 2\theta)^{-1/2} \frac{d}{d\theta} (\csc 2\theta) = -\frac{1}{2} (\csc 2\theta)^{-1/2} (\csc 2\theta \cot 2\theta)(2) = -\sqrt{\csc 2\theta} \cot 2\theta$$

11. $f(x) = x^2 \sin x$: $f'(x) = x^2 \frac{d}{dx} (\sin x) + \sin x \frac{d}{dx} (x^2) = x^2 \cos x + 2x \sin x$

12. $f(x) = \frac{\cos x}{x}$: $f'(x) = \frac{x \frac{d}{dx} (\cos x) - \cos x \frac{d}{dx} (x)}{x^2} = \frac{-x \sin x - \cos x}{x^2}$

13. Let $y = x \sin x$; find y''' .

$$y' = x \cos x + \sin x$$

$$y'' = x(-\sin x) + \cos x + \cos x = -x \sin x + 2 \cos x$$

$$y''' = -x \cos x - \sin x - 2 \sin x = -x \cos x - 3 \sin x$$

14. Let $y = \tan^2 (3x - 2)$; find y'' .

$$y' = 2 \tan (3x - 2) \sec^2 (3x - 2) \cdot 3 = 6 \tan (3x - 2) \sec^2 (3x - 2)$$

$$y'' = 6 [\tan (3x - 2) \cdot 2 \sec (3x - 2) \cdot \sec (3x - 2) \tan (3x - 2) \cdot 3 + \sec^2 (3x - 2) \sec^2 (3x - 2) \cdot 3] \\ = 36 \tan^2 (3x - 2) \sec^2 (3x - 2) + 18 \sec^4 (3x - 2)$$

15. Let $y = \sin (x + y)$; find y' .

$$y' = \cos (x + y) \cdot (1 + y'), \quad \text{so that} \quad y' = \frac{\cos (x + y)}{1 - \cos (x + y)}$$

16. Let $\sin y + \cos x = 1$; find y'' .

$$\cos y \cdot y' - \sin x = 0. \quad \text{So} \quad y' = \frac{\sin x}{\cos y}$$

Then
$$y'' = \frac{\cos y \cos x - \sin x (-\sin y) \cdot y'}{\cos^2 y} = \frac{\cos x \cos y + \sin x \sin y \cdot y'}{\cos^2 y} \\ = \frac{\cos x \cos y + \sin x \sin y (\sin x) / (\cos y)}{\cos^2 y} = \frac{\cos x \cos^2 y + \sin^2 x \sin y}{\cos^3 y}$$

17. Find $f'(\pi/3)$, $f''(\pi/3)$, and $f'''(\pi/3)$, given $f(x) = \sin x \cos 3x$.

$$f'(x) = -3 \sin x \sin 3x + \cos 3x \cos x \\ = (\cos 3x \cos x - \sin 3x \sin x) - 2 \sin x \sin 3x \\ = \cos 4x - 2 \sin x \sin 3x$$

So $f'(\pi/3) = -\frac{1}{2} - 2(\sqrt{3}/2)(0) = -\frac{1}{2}$

$$f''(x) = -4 \sin 4x - 2(3 \sin x \cos 3x + \sin 3x \cos x) \\ = -4 \sin 4x - 2(\sin x \cos 3x + \sin 3x \cos x) - 4 \sin x \cos 3x \\ = -6 \sin 4x - 4f(x)$$

So $f''(\pi/3) = -6(-\sqrt{3}/2) - 4(\sqrt{3}/2)(-1) = 5\sqrt{3}$

$$f'''(x) = -24 \cos 4x - 4f'(x). \quad \text{So} \quad f'''(\pi/3) = -24(-\frac{1}{2}) - 4(-\frac{1}{2}) = 14$$

18. Find the acute angles of intersection of the curves (1) $y = 2 \sin^2 x$ and (2) $y = \cos 2x$ on the interval $0 < x < 2\pi$. (See Fig. 17-6.)

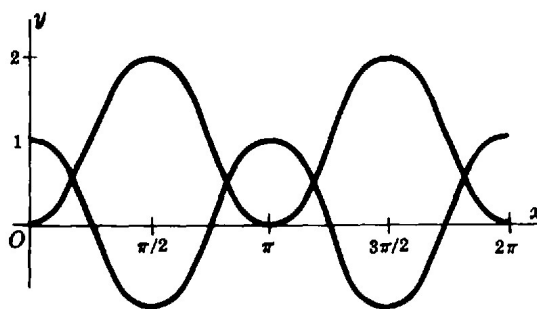


Fig. 17-6

We solve $2 \sin^2 x = \cos 2x = 1 - 2 \sin^2 x$ to obtain $\pi/6, 5\pi/6, 7\pi/6$, and $11\pi/6$ as the abscissas of the points of intersection.

Moreover, $y' = 4 \sin x \cos x$ for (1), and $y' = -2 \sin 2x$ for (2). Hence, at the point $\pi/6$, the curves have slopes $m_1 = \sqrt{3}$ and $m_2 = -\sqrt{3}$, respectively.

Since $\tan \phi = \frac{\sqrt{3} + \sqrt{3}}{1 - 3} = -\sqrt{3}$, the acute angle of intersection is 60° . At each of the remaining intersection points, the acute angle of intersection is also 60° .

19. A rectangular plot of ground has two adjacent sides along Highways 20 and 32. In the plot is a small lake, one end of which is 256 ft from Highway 20 and 108 ft from Highway 32 (see Fig. 17-7). Find the length of the shortest straight path which cuts across the plot from one highway to the other and touches the end of the lake.

Let s be the length of the path, and θ the angle it makes with Highway 32. Then

$$s = AP + PB = 108 \csc \theta + 256 \sec \theta$$

$$\frac{ds}{d\theta} = -108 \csc \theta \cot \theta + 256 \sec \theta \tan \theta = \frac{-108 \cos^3 \theta + 256 \sin^3 \theta}{\sin^2 \theta \cos^2 \theta}$$

Now $ds/d\theta = 0$ when $-108 \cos^3 \theta + 256 \sin^3 \theta = 0$, or when $\tan^3 \theta = 27/64$, and the critical value is $\theta = \arctan 3/4$. Then $s = 108 \csc \theta + 256 \sec \theta = 108(5/3) + 256(5/4) = 500$ ft.

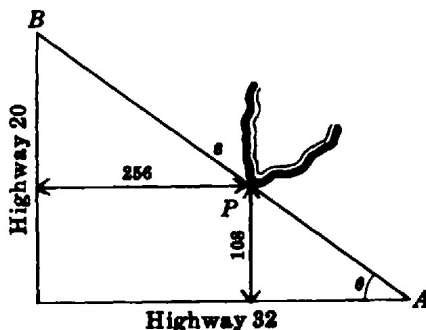


Fig. 17-7

20. Discuss the curve $y = f(x) = 4 \sin x - 3 \cos x$ on the interval $[0, 2\pi]$.

When $x = 0$, $y = f(0) = 4(0) - 3(1) = -3$.

Setting $f(x) = 0$ gives $\tan x = 3/4$, and the x -intercepts are $x = 0.64$ rad and $x = \pi + 0.64 = 3.78$ rad.
 $f'(x) = 4 \cos x + 3 \sin x$. Setting $f'(x) = 0$ gives $\tan x = -\frac{4}{3}$, and the critical values are $x = \pi - 0.93 = 2.21$ and $x = 2\pi - 0.93 = 5.35$.

$f''(x) = -4 \sin x + 3 \cos x$. Setting $f''(x) = 0$ gives $\tan x = 3/4$, and the possible points of inflection are $x = 0.64$ and $x = \pi + 0.64 = 3.78$.

$f'''(x) = -4 \cos x - 3 \sin x$. In addition,

1. When $x = 2.21$, $\sin x = 4/5$ and $\cos x = -3/5$; then $f''(x) < 0$, so $x = 2.21$ yields a relative maximum of 5. $x = 5.35$ yields a relative minimum of -5 .
2. $f'''(0.64) \neq 0$ and $f'''(3.78) \neq 0$. The points of inflection are $(0.64, 0)$ and $(3.78, 0)$.
3. The curve is concave upward from $x = 0$ to $x = 0.64$; concave downward from $x = 0.64$ to 3.78 ; and concave upward from $x = 3.78$ to 2π . (See Fig. 17-8.)

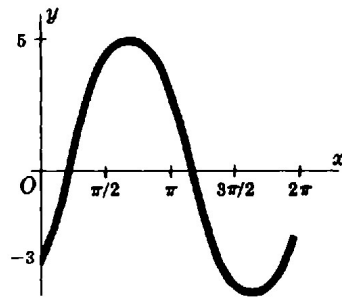


Fig. 17-8

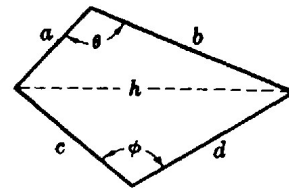


Fig. 17-9

21. Four bars of lengths a , b , c , and d are hinged together to form a quadrilateral (Fig. 17-9). Show that its area A is greatest when the opposite angles are supplementary.

Denote by θ the angle included by the bars of lengths a and b , by ϕ the opposite angle, and by h the length of the diagonal opposite these angles. We are required to maximize

$$A = \frac{1}{2}ab \sin \theta + \frac{1}{2}cd \sin \phi$$

subject to

$$h^2 = a^2 + b^2 - 2ab \cos \theta = c^2 + d^2 - 2cd \cos \phi$$

Differentiation with respect to θ yields, respectively,

$$\frac{dA}{d\theta} = \frac{1}{2}ab \cos \theta + \frac{1}{2}cd \cos \phi \frac{d\phi}{d\theta} = 0 \quad \text{and} \quad ab \sin \theta = cd \sin \phi \frac{d\phi}{d\theta}$$

We solve for $d\phi/d\theta$ in the second of these equations and substitute in the first to obtain

$$ab \cos \theta + cd \cos \phi \frac{ab \sin \theta}{cd \sin \phi} = 0 \quad \text{or} \quad \sin \phi \cos \theta + \cos \phi \sin \theta = \sin(\phi + \theta) = 0$$

Then $\phi + \theta = 0$ or π , the first of which is easily rejected.

22. A bombardier is sighting on a target on the ground directly ahead. If the bomber is flying 2 mi above the ground at 240 mi/h, how fast must the sighting instrument be turning when the angle between the path of the bomber and the line of sight is 30° ?

We have $dx/dt = -240$ mi/h, $\theta = 30^\circ$, and $x = 2 \cot \theta$ in Fig. 17-10. From the last equation,

$$\frac{dx}{dt} = -2 \csc^2 \theta \frac{d\theta}{dt} \quad \text{or} \quad -240 = -2(4) \frac{d\theta}{dt} \quad \text{so} \quad \frac{d\theta}{dt} = 30 \text{ rad/h} = \frac{3}{2\pi} \text{ degree/sec}$$

23. A ray of light passes through the air with velocity v_1 from a point P , a units above the surface of a body of water, to some point O on the surface and then with velocity v_2 to a point Q , b

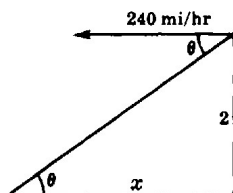


Fig. 17-10

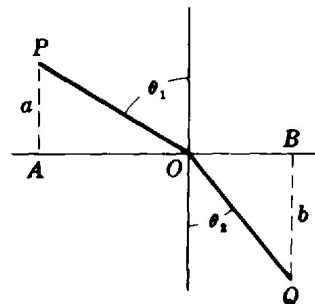


Fig. 17-11

units below the surface (Fig. 17-11). If OP and OQ make angles of θ_1 and θ_2 with a perpendicular to the surface, show that passage from P to Q is most rapid when $\sin \theta_1 / \sin \theta_2 = v_1 / v_2$.

Let t denote the time required for passage from P to Q , and c the distance from A to B ; then

$$t = \frac{a \sec \theta_1}{v_1} + \frac{b \sec \theta_2}{v_2} \quad \text{and} \quad c = a \tan \theta_1 + b \tan \theta_2$$

Differentiating with respect to θ_1 yields

$$\frac{dt}{d\theta_1} = \frac{a \sec \theta_1 \tan \theta_1}{v_1} + \frac{b \tan \theta_2 \sec \theta_2}{v_2} \frac{d\theta_2}{d\theta_1} \quad \text{and} \quad 0 = a \sec^2 \theta_1 + b \sec^2 \theta_2 \frac{d\theta_2}{d\theta_1}$$

From the last equation, $\frac{d\theta_2}{d\theta_1} = -\frac{a \sec^2 \theta_1}{b \sec^2 \theta_2}$. For t to be a minimum, it is necessary that

$$\frac{dt}{d\theta_1} = \frac{a \sec \theta_1 \tan \theta_1}{v_1} + \frac{b \sec \theta_2 \tan \theta_2}{v_2} \left(-\frac{a \sec^2 \theta_1}{b \sec^2 \theta_2} \right) = 0$$

from which the required relation follows.

Supplementary Problems

24. Evaluate: (a) $\lim_{x \rightarrow 0} \frac{\sin 2x}{x} = 2 \lim_{x \rightarrow 0} \frac{\sin 2x}{2x}$; (b) $\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx}$; (c) $\lim_{x \rightarrow 0} \frac{\sin^3 2x}{x \sin^2 3x}$.

Ans. (a) 2; (b) a/b ; (c) $8/9$

25. Derive differentiation formula 17, using first (a) $\cot u = \frac{\cos u}{\sin u}$ and then (b) $\cot u = \frac{1}{\tan u}$. Also derive differentiation formulas 18 and 19.

In Problems 26 to 45, find the derivative dy/dx or $dp/d\theta$.

- | | | | | | | | |
|-----|--|------|---|-----|---------------------------|------|------------------------|
| 26. | $y = 3 \sin 2x$ | Ans. | $6 \cos 2x$ | 27. | $y = 4 \cos \frac{1}{2}x$ | Ans. | $-2 \sin \frac{1}{2}x$ |
| 28. | $y = 4 \tan 5x$ | Ans. | $20 \sec^2 5x$ | 29. | $y = \frac{1}{4} \cot 8x$ | Ans. | $-2 \csc^2 8x$ |
| 30. | $y = 9 \sec \frac{1}{3}x$ | Ans. | $3 \sec \frac{1}{3}x \tan \frac{1}{3}x$ | 31. | $y = \frac{1}{4} \csc 4x$ | Ans. | $-\csc 4x \cot 4x$ |
| 32. | $y = \sin x - x \cos x + x^2 + 4x + 3$ | Ans. | $x \sin x + 2x + 4$ | | | | |
| 33. | $\rho = \sqrt{\sin \theta}$ | Ans. | $(\cos \theta) / (2\sqrt{\sin \theta})$ | 34. | $y = \sin 2/x$ | Ans. | $(-2 \cos 2/x) / x^2$ |

35. $y = \cos(1 - x^2)$ *Ans.* $2x \sin(1 - x^2)$
36. $y = \cos(1 - x)^2$ *Ans.* $2(1 - x) \sin(1 - x)^2$
37. $y = \sin^2(3x - 2)$ *Ans.* $3 \sin(6x - 4)$
38. $y = \sin^3(2x - 3)$ *Ans.* $-\frac{3}{2} \{\cos(6x - 9) - \cos(2x - 3)\}$
39. $y = \frac{1}{2} \tan x \sin 2x$ *Ans.* $\sin 2x$
40. $\rho = \frac{1}{(\sec 2\theta - 1)^{3/2}}$ *Ans.* $\frac{-3 \sec 2\theta \tan 2\theta}{(\sec 2\theta - 1)^{5/2}}$
41. $\rho = \frac{\tan 2\theta}{1 - \cot 2\theta}$ *Ans.* $2 \frac{\sec^2 2\theta - 4 \csc 4\theta}{(1 - \cot 2\theta)^2}$
42. $y = x^2 \sin x + 2x \cos x - 2 \sin x$ *Ans.* $x^2 \cos x$
43. $\sin y = \cos 2x$ *Ans.* $-2 \sin 2x / \cos y$
44. $\cos 3y = \tan 2x$ *Ans.* $-2 \sec^2 2x / 3 \sin 3y$
45. $x \cos y = \sin(x + y)$ *Ans.* $\frac{\cos y - \cos(x + y)}{x \sin y + \cos(x + y)}$
46. If $x = A \sin kt + B \cos kt$ for A , B , and k constants, show that $\frac{d^2x}{dt^2} = -k^2x$ and $\frac{d^{2n}x}{dt^{2n}} = (-1)^n k^{2n}x$.
47. Show: (a) $y'' + 4y = 0$ when $y = 3 \sin(2x + 3)$; (b) $y''' + y'' + y' + y = 0$ when $y = \sin x + 2 \cos x$.
48. Discuss and sketch on the interval $0 \leq x < 2\pi$:
 (a) $y = \frac{1}{2} \sin 2x$ (b) $y = \cos^2 x - \cos x$ (c) $y = x - 2 \sin x$
 (d) $y = \sin x (1 + \cos x)$ (e) $y = 4 \cos^3 x - 3 \cos x$
- Ans.* (a) maximum at $x = \pi/4, 5\pi/4$; minimum at $x = 3\pi/4, 7\pi/4$; inflection point at $x = 0, \pi/2, \pi, 3\pi/2$
 (b) maximum at $x = 0, \pi$; minimum at $x = \pi/3, 5\pi/3$; inflection point at $x = 32^\circ 32', 126^\circ 23', 233^\circ 37', 327^\circ 28'$
 (c) maximum at $x = 5\pi/3$; minimum at $x = \pi/3$; inflection point at $x = 0, \pi$
 (d) maximum at $x = \pi/3$; minimum at $x = 5\pi/3$; inflection point at $x = 0, \pi, 104^\circ 29', 255^\circ 31'$
 (e) maximum at $x = 0, 2\pi/3, 4\pi/3$; minimum at $x = \pi/3, \pi, 5\pi/3$; inflection point at $x = \pi/2, 3\pi/2, \pi/6, 5\pi/6, 7\pi/6, 11\pi/6$
49. If the angle of elevation of the sun is 45° and is decreasing at $\frac{1}{4}$ rad/h, how fast is the shadow cast on level ground by a pole 50 ft tall lengthening? *Ans.* 25 ft/h
50. A kite, 120 ft above the ground, is moving horizontally at the rate of 10 ft/sec. At what rate is the inclination of the string to the horizontal diminishing when 240 ft of string are paid out?
Ans. $\frac{1}{48}$ rad/sec
51. A revolving beacon is situated 3600 ft off a straight shore. If the beacon turns at 4π rad/min, how fast does the beam sweep along the shore at (a) its nearest point, (b) at a point 4800 ft from the nearest point? *Ans.* (a) 240π ft/sec; (b) $2000\pi/3$ ft/sec
52. Two sides of a triangle are 15 and 20 ft long, respectively. (a) How fast is the third side increasing if the angle between the given sides is 60° and is increasing at the rate $2^\circ/\text{sec}$? (b) How fast is the area increasing? *Ans.* (a) $\pi/\sqrt{39}$ ft/sec; (b) $\frac{5}{6}\pi$ ft²/sec

Chapter 18

Differentiation of Inverse Trigonometric Functions

THE INVERSE TRIGONOMETRIC FUNCTIONS. If $x = \sin y$, the inverse function is written $y = \arcsin x$. (An alternative notation is $y = \sin^{-1} x$.) The domain of $\arcsin x$ is $-1 \leq x \leq 1$, which is the range of $\sin y$. The range of $\arcsin x$ is the set of real numbers, which is the domain of $\sin y$. The domain and range of the remaining inverse trigonometric functions may be established in a similar manner.

The inverse trigonometric functions are multivalued. In order that there be agreement on separating the graph into single-valued arcs, we define in Table 18-1 one such arc (called the *principal branch*) for each function. In Fig. 18-1, the principal branches are indicated by a thicker curve.

Table 18-1

Function	Principal Branch
$y = \arcsin x$	$-\frac{1}{2}\pi \leq y \leq \frac{1}{2}\pi$
$y = \arccos x$	$0 \leq y \leq \pi$
$y = \arctan x$	$-\frac{1}{2}\pi < y < \frac{1}{2}\pi$
$y = \operatorname{arccot} x$	$0 < y < \pi$
$y = \operatorname{arcsec} x$	$-\pi \leq y < -\frac{1}{2}\pi, 0 \leq y < \frac{1}{2}\pi$
$y = \operatorname{arccsc} x$	$-\pi < y \leq -\frac{1}{2}\pi, 0 < y \leq \frac{1}{2}\pi$

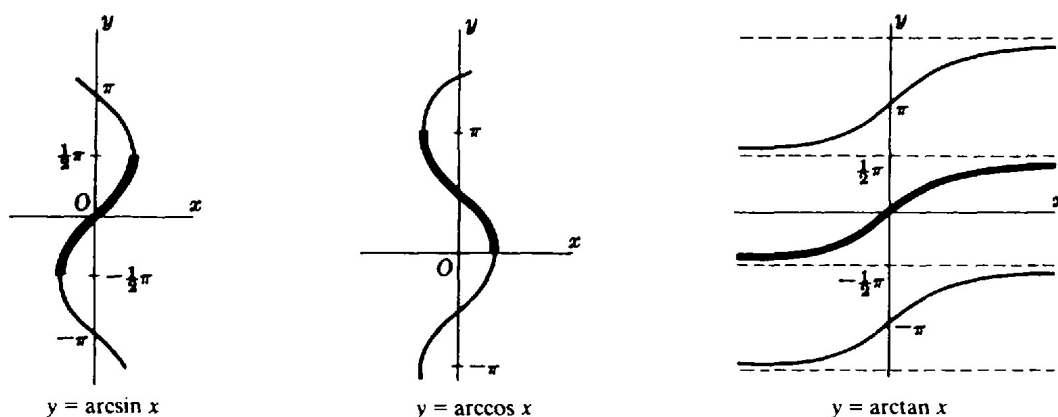


Fig. 18-1

DIFFERENTIATION FORMULAS

$$20. \quad \frac{d}{dx} (\arcsin x) = \frac{1}{\sqrt{1-x^2}}$$

$$21. \quad \frac{d}{dx} (\arccos x) = -\frac{1}{\sqrt{1-x^2}}$$

$$22. \quad \frac{d}{dx} (\arctan x) = \frac{1}{1+x^2}$$

$$23. \quad \frac{d}{dx} (\operatorname{arccot} x) = -\frac{1}{1+x^2}$$

$$24. \quad \frac{d}{dx} (\operatorname{arcsec} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$25. \quad \frac{d}{dx} (\operatorname{arccsc} x) = -\frac{1}{x\sqrt{x^2-1}}$$

Solved Problems

1. Derive: (a) $\frac{d}{dx} (\arcsin x) = \frac{1}{\sqrt{1-x^2}}$; (b) $\frac{d}{dx} (\operatorname{arcsec} x) = \frac{1}{x\sqrt{x^2-1}}$.

(a) Let $y = \arcsin x$. Then $x = \sin y$ and

$$1 = \frac{d}{dx} (x) = \frac{d}{dx} (\sin y) = \frac{d}{dy} (\sin y) \frac{dy}{dx} = \cos y \frac{dy}{dx} = \sqrt{1-x^2} \frac{dy}{dx}$$

the sign being positive since $\cos y \geq 0$ on the interval $-\frac{1}{2}\pi \leq y \leq \frac{1}{2}\pi$. Thus, $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$.

(b) Let $y = \operatorname{arcsec} x$. Then $x = \sec y$ and

$$1 = \frac{d}{dx} (x) = \frac{d}{dx} (\sec y) = \frac{d}{dy} (\sec y) \frac{dy}{dx} = \sec y \tan y \frac{dy}{dx} = x\sqrt{x^2-1} \frac{dy}{dx}$$

the sign being positive since $\tan y \geq 0$ on the intervals $0 \leq y < \frac{1}{2}\pi$ and $-\pi \leq y < -\frac{1}{2}\pi$. Thus, $\frac{d}{dx} (\operatorname{arcsec} x) = \frac{1}{x\sqrt{x^2-1}}$.

In Problems 2 to 8, find the first derivative.

2. $y = \arcsin (2x-3)$: $\frac{dy}{dx} = \frac{1}{\sqrt{1-(2x-3)^2}} \frac{d}{dx} (2x-3) = \frac{1}{\sqrt{3x-x^2-2}}$

3. $y = \arccos x^2$: $\frac{dy}{dx} = -\frac{1}{\sqrt{1-x^4}} \frac{d}{dx} (x^2) = -\frac{2x}{\sqrt{1-x^4}}$

4. $y = \arctan 3x^2$: $\frac{dy}{dx} = \frac{1}{1+(3x^2)^2} \frac{d}{dx} (3x^2) = \frac{6x}{1+9x^4}$

5. $f(x) = \operatorname{arccot} \frac{1+x}{1-x}$:

$$f'(x) = -\frac{1}{1+\left(\frac{1+x}{1-x}\right)^2} \frac{d}{dx} \left(\frac{1+x}{1-x}\right) = -\frac{1}{1+\left(\frac{1+x}{1-x}\right)^2} \frac{(1-x)-(1+x)(-1)}{(1-x)^2} = -\frac{1}{1+x^2}$$

6. $f(x) = x\sqrt{a^2-x^2} + a^2 \arcsin \frac{x}{a}$:

$$f'(x) = x[\frac{1}{2}(a^2-x^2)^{-1/2}(-2x)] + (a^2-x^2)^{1/2} + a^2 \frac{1}{\sqrt{1-(x/a)^2}} \frac{1}{a} = 2\sqrt{a^2-x^2}$$

7. $y = x \operatorname{arccsc} \frac{1}{x} + \sqrt{1-x^2}$:

$$y' = x \left[\frac{-1}{\frac{1}{x} \sqrt{\frac{1}{x^2}-1}} \frac{d}{dx} \left(\frac{1}{x}\right) \right] + \operatorname{arccsc} \frac{1}{x} \frac{d}{dx} (x) + \frac{1}{2}(1-x^2)^{-1/2}(-2x) = \operatorname{arccsc} \frac{1}{x}$$

8. $y = \frac{1}{ab} \arctan \left(\frac{b}{a} \tan x \right)$:

$$y' = \frac{1}{ab} \left[\frac{1}{1 + \left(\frac{b}{a} \tan x\right)^2} \frac{d}{dx} \left(\frac{b}{a} \tan x\right) \right] = \frac{1}{ab} \frac{a^2}{a^2 + b^2 \tan^2 x} \frac{b}{a} \sec^2 x$$

$$= \frac{\sec^2 x}{a^2 + b^2 \tan^2 x} = \frac{1}{a^2 \cos^2 x + b^2 \sin^2 x}$$

9. $y^2 \sin x + y = \arctan x$; find y' .

$$2yy' \sin x + y^2 \cos x + y' = \frac{1}{1+x^2}$$

Hence, $y'(2y \sin x + 1) = \frac{1}{1+x^2} - y^2 \cos x$ and $y' = \frac{1 - (1+x^2)y^2 \cos x}{(1+x^2)(2y \sin x + 1)}$

10. In a circular arena (Fig. 18-2) there is a light at L . A boy starting from B runs at the rate of 10 ft/sec toward the center O . At what rate will his shadow be moving along the side when he is halfway from B to O ?

Let P , a point x feet from B , be the position of the boy at time t ; denote by r the radius of the arena, by θ the angle OLP , and by s the arc intercepted by θ . Then $s = r(2\theta)$, and $\theta = \arctan OP/LO = \arctan (r-x)/r$. Hence,

$$\frac{ds}{dt} = 2r \frac{d\theta}{dt} = 2r \frac{1}{1 + [(r-x)/r]^2} \left(-\frac{1}{r}\right) \frac{dx}{dt} = \frac{-2r^2}{x^2 - 2rx + 2r^2} \frac{dx}{dt}$$

When $x = \frac{1}{2}r$ and $dx/dt = 10$, $ds/dt = -16$ ft/sec. The shadow is moving along the wall at 16 ft/sec.

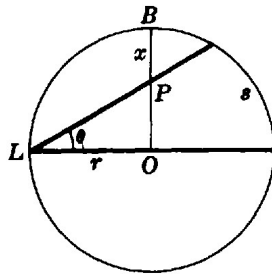


Fig. 18-2

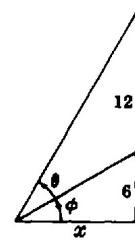


Fig. 18-3

11. The lower edge of a mural, 12 ft high, is 6 ft above an observer's eyes. Under the assumption that the most favorable view is obtained when the angle subtended by the mural at the eye is a maximum, at what distance from the wall should the observer stand?

Let θ denote the subtended angle, and x the distance from the wall. From Fig. 18-3, $\tan(\theta + \phi) = 18/x$, $\tan \phi = 6/x$, and

$$\tan \theta = \tan[(\theta + \phi) - \phi] = \frac{\tan(\theta + \phi) - \tan \phi}{1 + \tan(\theta + \phi) \tan \phi} = \frac{18/x - 6/x}{1 + (18/x)(6/x)} = \frac{12x}{x^2 + 108}$$

Then $\theta = \arctan \frac{12x}{x^2 + 108}$ and $\frac{d\theta}{dx} = \frac{12(-x^2 + 108)}{x^4 + 360x^2 + 11,664}$. The critical value is $x = 6\sqrt{3} \sim 10.4$. The observer should stand 10.4 ft in front of the wall.

Supplementary Problems

12. Derive differentiation formulas 21, 22, 23, and 25.

In Problems 13 to 20, find dy/dx .

13. $y = \arcsin 3x$ *Ans.* $\frac{3}{\sqrt{1-9x^2}}$ 14. $y = \arccos \frac{1}{2}x$ *Ans.* $-\frac{1}{\sqrt{4-x^2}}$

15. $y = \arctan \frac{3}{x}$ *Ans.* $-\frac{3}{x^2+9}$ 16. $y = \arcsin (x-1)$ *Ans.* $\frac{1}{\sqrt{2x-x^2}}$

17. $y = x^2 \arccos 2/x$ *Ans.* $2x \left(\arccos \frac{2}{x} + \frac{1}{\sqrt{x^2-4}} \right)$

18. $y = \frac{x}{\sqrt{a^2-x^2}} - \arcsin \frac{x}{a}$ *Ans.* $\frac{x^2}{(a^2-x^2)^{3/2}}$

19. $y = (x-a)\sqrt{2ax-x^2} + a^2 \arcsin \frac{x-a}{a}$ *Ans.* $2\sqrt{2ax-x^2}$

20. $y = \frac{\sqrt{x^2-4}}{x^2} + \frac{1}{2} \operatorname{arcsec} \frac{x}{2}$ *Ans.* $\frac{8}{x^3\sqrt{x^2-4}}$

21. A light is to be placed directly above the center of a circular plot of radius 30 ft, at such a height that the edge of the plot will get maximum illumination. Find the height if the intensity at any point on the edge is directly proportional to the cosine of the angle of incidence (angle between the ray of light and the vertical) and inversely proportional to the square of the distance from the source. (*Hint:* Let x be the required height, y the distance from the light to a point on the edge, and θ the angle of incidence. Then

$$I = k \frac{\cos \theta}{y^2} = \frac{kx}{(x^2+900)^{3/2}}. \quad \text{Ans. } 15\sqrt{2} \text{ ft}$$

22. Two ships sail from A at the same time. One sails south at 15 mi/h; the other sails east at 25 mi/h for 1 h and then turns north. Find the rate of rotation of the line joining them after 3 h. *Ans.* $\frac{20}{193} \text{ rad/h}$

Chapter 19

Differentiation of Exponential and Logarithmic Functions

DEFINE THE NUMBER e by the equation

$$e = \lim_{h \rightarrow +\infty} \left(1 + \frac{1}{h}\right)^h$$

Then e also can be represented by $\lim_{k \rightarrow 0} (1+k)^{1/k}$. In addition, it can be shown that

$$e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} + \cdots = 2.71828 \dots$$

The number e will serve as a base for the natural logarithm function (See Problem 1.)

LOGARITHMIC FUNCTIONS. Assume $a > 0$ and $a \neq 1$. If $a^y = x$, then define $y = \log_a x$. Another definition of $\log_a x$ will be given in Chapter 40.

NOTATION. Let $\ln x \equiv \log_e x$. (Then $\ln x$ is called the *natural logarithm* of x .) See also Fig. 19-1.
Let $\log x \equiv \log_{10} x$.

The domain of $\log_a x$ is $x > 0$; the range is the set of real numbers.

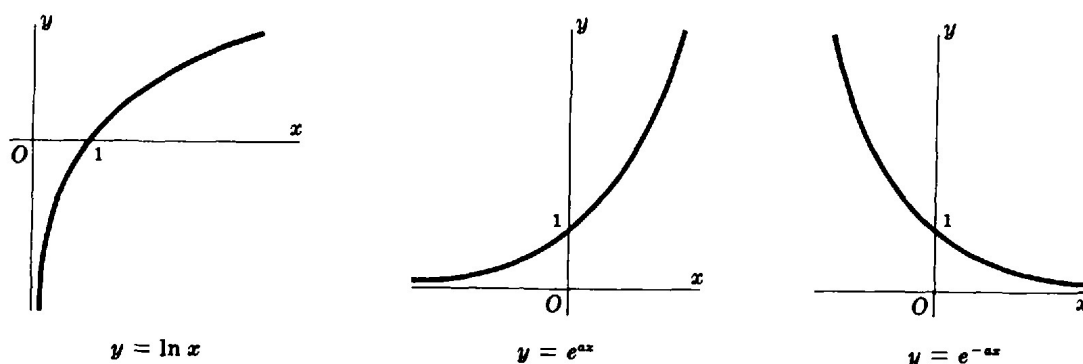


Fig. 19-1

DIFFERENTIATION FORMULAS

- | | |
|---|--|
| 26. $\frac{d}{dx} (\log_a x) = \frac{1}{x} \log_a e, a > 0, a \neq 1$ | 27. $\frac{d}{dx} (\ln x) = \frac{1}{x}$ |
| 28. $\frac{d}{dx} (a^x) = a^x \ln a, a > 0$ | 29. $\frac{d}{dx} (e^x) = e^x$ |

(See Problems 2 to 17.)

LOGARITHMIC DIFFERENTIATION. If a differentiable function $y = f(x)$ is the product and/or quotient of several factors, the process of differentiation may be simplified by taking the natural

logarithm of the function before differentiation. This amounts to using the formula

$$30. \quad \frac{d}{dx} (y) = y \frac{d}{dx} (\ln y)$$

(See Problems 18 and 19.)

BASIC PROPERTIES OF LOGARITHMS

Property 19.1: $\log_a 1 = 0$ (In particular, $\ln 1 = 0$.)

Property 19.2: $\log_a a = 1$ (In particular, $\ln e = 1$.)

Property 19.3: $\log_a uv = \log_a u + \log_a v$

Property 19.4: $\log_a \frac{u}{v} = \log_a u - \log_a v$

Property 19.5: $\log_a u^r = r \log_a u$

Solved Problems

1. Verify: $2 < \lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^n < 3$.

By the binomial theorem, for n a positive integer,

$$\begin{aligned} \left(1 + \frac{1}{n}\right)^n &= 1 + n \frac{1}{n} + \frac{n(n-1)}{1 \cdot 2} \left(\frac{1}{n}\right)^2 + \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3} \left(\frac{1}{n}\right)^3 + \cdots + \frac{n(n-1)(n-2) \cdots 1}{1 \cdot 2 \cdot 3 \cdots n} \left(\frac{1}{n}\right)^n \\ &= 1 + 1 + \left(1 - \frac{1}{n}\right) \frac{1}{2!} + \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \frac{1}{3!} + \cdots + \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{n-1}{n}\right) \frac{1}{n!} \quad (1) \end{aligned}$$

Clearly, for every value of $n \neq 1$, $\left(1 + \frac{1}{n}\right)^n > 2$. Also, if in (1) each difference $\left(1 - \frac{1}{n}\right)$, $\left(1 - \frac{2}{n}\right)$, $\dots$ is replaced by the larger number 1, we have

$$\begin{aligned} \left(1 + \frac{1}{n}\right)^n &< 2 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} \\ &< 2 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^{n-1}} \quad \left(\text{since } \frac{1}{n!} < \frac{1}{2^{n-1}}\right) \\ &< 3 \quad \left(\text{since } \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots + \frac{1}{2^{n-1}} < 1\right) \end{aligned}$$

Hence, $2 < \left(1 + \frac{1}{n}\right)^n < 3$.

Let $n \rightarrow \infty$ through positive integer values; then

$$1 - \frac{1}{n} \rightarrow 1, \quad 1 - \frac{2}{n} \rightarrow 1, \dots, \quad \text{and} \quad \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k}{n}\right) \frac{1}{k!} \rightarrow \frac{1}{k!}$$

This suggests that $\lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^n = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{k!} + \cdots = 2.71828 \dots$

2. Derive $\frac{d}{dx} (\log_a x) = \frac{1}{x} \log_a e$ and $\frac{d}{dx} (\ln x) = \frac{1}{x}$.

Let $y = \log_a x$. Then

$$y + \Delta y = \log_a (x + \Delta x)$$

$$\Delta y = \log_a (x + \Delta x) - \log_a x = \log_a \frac{x + \Delta x}{x} = \log_a \left(1 + \frac{\Delta x}{x}\right)$$

$$\frac{\Delta y}{\Delta x} = \frac{1}{\Delta x} \log_a \left(1 + \frac{\Delta x}{x}\right) = \frac{1}{x} \frac{x}{\Delta x} \log_a \left(1 + \frac{\Delta x}{x}\right) = \frac{1}{x} \log_a \left(1 + \frac{\Delta x}{x}\right)^{x/\Delta x}$$

and $\frac{dy}{dx} = \frac{1}{x} \lim_{\Delta x \rightarrow 0} \log_a \left(1 + \frac{\Delta x}{x}\right)^{x/\Delta x} = \frac{1}{x} \log_a \left[\lim_{\Delta x \rightarrow 0} \left(1 + \frac{\Delta x}{x}\right)^{x/\Delta x} \right] = \frac{1}{x} \log_a e$

When $a = e$, $\log_a e = \log_e e = 1$ and $\frac{d}{dx} (\ln x) = \frac{1}{x}$.

In Problems 3 to 9, find the first derivative.

3. $y = \log_a (3x^2 - 5):$ $\frac{dy}{dx} = \frac{1}{3x^2 - 5} (\log_a e) \frac{d}{dx} (3x^2 - 5) = \frac{6x}{3x^2 - 5} \log_a e$

4. $y = \ln (x + 3)^2 = 2 \ln (x + 3):$ $\frac{dy}{dx} = 2 \frac{1}{x + 3} \frac{d}{dx} (x + 3) = \frac{2}{x + 3}$

5. $y = \ln^2 (x + 3):$
 $y' = 2 \ln (x + 3) \frac{d}{dx} [\ln (x + 3)] = 2 \ln (x + 3) \frac{1}{x + 3} \frac{d}{dx} (x + 3) = \frac{2 \ln (x + 3)}{x + 3}$

6. $y = \ln (x^3 + 2)(x^2 + 3) = \ln (x^3 + 2) + \ln (x^2 + 3):$
 $y' = \frac{1}{x^3 + 2} \frac{d}{dx} (x^3 + 2) + \frac{1}{x^2 + 3} \frac{d}{dx} (x^2 + 3) = \frac{3x^2}{x^3 + 2} + \frac{2x}{x^2 + 3}$

7. $f(x) = \ln \frac{x^4}{(3x - 4)^2} = \ln x^4 - \ln (3x - 4)^2 = 4 \ln x - 2 \ln (3x - 4):$
 $f'(x) = 4 \frac{1}{x} \frac{d}{dx} (x) - 2 \frac{1}{3x - 4} \frac{d}{dx} (3x - 4) = \frac{4}{x} - \frac{6}{3x - 4}$

8. $y = \ln \sin 3x:$ $y' = \frac{1}{\sin 3x} \frac{d}{dx} (\sin 3x) = 3 \frac{\cos 3x}{\sin 3x} = 3 \cot 3x$

9. $y = \ln (x + \sqrt{1 + x^2}):$
 $y' = \frac{1 + \frac{1}{2}(1 + x^2)^{-1/2}(2x)}{x + (1 + x^2)^{1/2}} = \frac{1 + x(1 + x^2)^{-1/2}}{x + (1 + x^2)^{1/2}} \frac{(1 + x^2)^{1/2}}{(1 + x^2)^{1/2}} = \frac{1}{\sqrt{1 + x^2}}$

10. Derive $\frac{d}{dx} (a^x) = (\ln a) a^x$ and $\frac{d}{dx} (e^x) = e^x$.

Let $y = a^x$. Then $\ln y = x \ln a$ and

$$\frac{d}{dx} (\ln y) = \frac{1}{y} \frac{dy}{dx} = \ln a \quad \text{or} \quad \frac{dy}{dx} = y \ln a = a^x \ln a$$

When $a = e$, $\ln a = \ln e = 1$ and we have $\frac{d}{dx} (e^x) = e^x$.

In Problems 11 to 15, find the first derivative.

$$11. \quad y = e^{-\frac{1}{2}x}; \quad y' = e^{-\frac{1}{2}x} \frac{d}{dx} \left(-\frac{1}{2}x \right) = -\frac{1}{2} e^{-\frac{1}{2}x}$$

$$12. \quad y = e^{x^2}; \quad y' = e^{x^2} \frac{d}{dx} (x^2) = 2xe^{x^2}$$

$$13. \quad y = a^{3x^2}; \quad y' = a^{3x^2} (\ln a) \frac{d}{dx} (3x^2) = 6xa^{3x^2} \ln a$$

$$14. \quad y = x^2 3^x; \quad y' = x^2 \frac{d}{dx} (3^x) + 3^x \frac{d}{dx} (x^2) = x^2 3^x \ln 3 + 3^x 2x = x 3^x (x \ln 3 + 2)$$

$$\begin{aligned} 15. \quad y = \frac{e^{ax} - e^{-ax}}{e^{ax} + e^{-ax}}; \quad y' &= \frac{(e^{ax} + e^{-ax}) \frac{d}{dx} (e^{ax} - e^{-ax}) - (e^{ax} - e^{-ax}) \frac{d}{dx} (e^{ax} + e^{-ax})}{(e^{ax} + e^{-ax})^2} \\ &= \frac{(e^{ax} + e^{-ax})(a)(e^{ax} + e^{-ax}) - (e^{ax} - e^{-ax})(a)(e^{ax} - e^{-ax})}{(e^{ax} + e^{-ax})^2} \\ &= a \frac{(e^{2ax} + 2 + e^{-2ax}) - (e^{2ax} - 2 + e^{-2ax})}{(e^{ax} + e^{-ax})^2} = \frac{4a}{(e^{ax} + e^{-ax})^2} \end{aligned}$$

$$16. \quad \text{Find } y'', \text{ given } y = e^{-x} \ln x.$$

$$\begin{aligned} y' &= e^{-x} \frac{d}{dx} (\ln x) + \ln x \frac{d}{dx} (e^{-x}) = \frac{e^{-x}}{x} - e^{-x} \ln x = \frac{e^{-x}}{x} - y \\ y'' &= \frac{x \frac{d}{dx} (e^{-x}) - e^{-x} \frac{d}{dx} (x)}{x^2} - y' = \frac{-xe^{-x} - e^{-x}}{x^2} - \frac{e^{-x}}{x} + e^{-x} \ln x = -e^{-x} \left(\frac{2}{x} + \frac{1}{x^2} - \ln x \right) \end{aligned}$$

$$17. \quad \text{Find } y'', \text{ given } y = e^{-2x} \sin 3x.$$

$$\begin{aligned} y' &= e^{-2x} \frac{d}{dx} (\sin 3x) + \sin 3x \frac{d}{dx} (e^{-2x}) = 3e^{-2x} \cos 3x - 2e^{-2x} \sin 3x = 3e^{-2x} \cos 3x - 2y \\ y'' &= 3e^{-2x} \frac{d}{dx} (\cos 3x) + 3 \cos 3x \frac{d}{dx} (e^{-2x}) - 2y' \\ &= -9e^{-2x} \sin 3x - 6e^{-2x} \cos 3x - 2(3e^{-2x} \cos 3x - 2e^{-2x} \sin 3x) \\ &= -e^{-2x} (12 \cos 3x + 5 \sin 3x) \end{aligned}$$

In Problems 18 and 19, use logarithmic differentiation to find the first derivative.

$$18. \quad y = (x^2 + 2)^3 (1 - x^3)^4$$

$$\ln y = \ln (x^2 + 2)^3 (1 - x^3)^4 = 3 \ln (x^2 + 2) + 4 \ln (1 - x^3)$$

$$\begin{aligned} y' &= y \frac{d}{dx} [3 \ln (x^2 + 2) + 4 \ln (1 - x^3)] = (x^2 + 2)^3 (1 - x^3)^4 \left(\frac{6x}{x^2 + 2} - \frac{12x^2}{1 - x^3} \right) \\ &= 6x(x^2 + 2)^2 (1 - x^3)^3 (1 - 4x - 3x^3) \end{aligned}$$

$$19. \quad y = \frac{x(1 - x^2)^2}{(1 + x^2)^{1/2}}$$

$$\ln y = \ln x + 2 \ln(1 - x^2) - \frac{1}{2} \ln(1 + x^2)$$

$$\begin{aligned} y' &= \frac{x(1-x^2)^2}{(1+x^2)^{1/2}} \left(\frac{1}{x} - \frac{4x}{1-x^2} - \frac{x}{1+x^2} \right) = \frac{(1-x^2)^2}{(1+x^2)^{1/2}} - \frac{4x^2(1-x^2)}{(1+x^2)^{1/2}} - \frac{x^2(1-x^2)^2}{(1+x^2)^{3/2}} \\ &= \frac{(1-5x^2-4x^4)(1-x^2)}{(1+x^2)^{3/2}} \end{aligned}$$

20. Locate (a) the relative maximum and minimum points and (b) the points of inflection of the curve $y = f(x) = x^2 e^x$ (Fig. 19-2).

$$f'(x) = 2xe^x + x^2 e^x = xe^x(2+x)$$

$$f''(x) = 2e^x + 4xe^x + x^2 e^x = e^x(2+4x+x^2)$$

$$f'''(x) = 6e^x + 6xe^x + x^2 e^x = e^x(6+6x+x^2)$$

- (a) Solving $f'(x) = 0$ gives the critical values $x = 0$ and $x = -2$. Then $f''(0) > 0$; so $(0, 0)$ is a relative minimum point. Also, $f''(-2) < 0$; so $(-2, 4/e^2)$ is a relative maximum point.
 (b) Solving $f''(x) = 0$ gives possible points of inflection at $x = -2 \pm \sqrt{2}$. Since $f'''(-2 - \sqrt{2}) \neq 0$ and $f'''(-2 + \sqrt{2}) \neq 0$, the points at $x = -2 \pm \sqrt{2}$ are points of inflection.

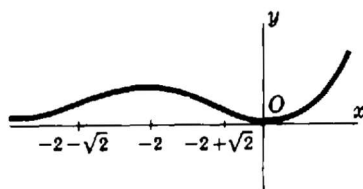


Fig. 19-2

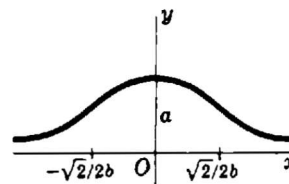


Fig. 19-3

21. Discuss the probability curve $y = ae^{-b^2 x^2}$, $a > 0$ (Fig. 19-3).

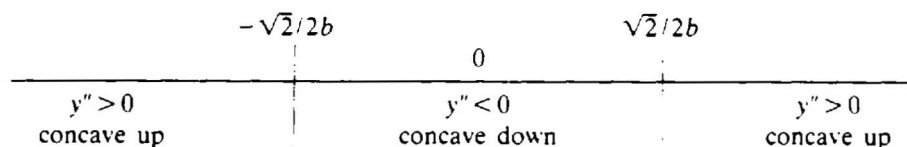
The curve lies entirely above the x axis, since $e^{-b^2 x^2} > 0$ for all x . As $x \rightarrow \pm\infty$, $y \rightarrow 0$; hence the x axis is a horizontal asymptote.

The first two derivatives are

$$y' = -2ab^2 x e^{-b^2 x^2} \quad \text{and} \quad y'' = 2ab^2(2b^2 x^2 - 1)e^{-b^2 x^2}$$

When $y' = 0$, $x = 0$, and when $x = 0$, $y'' < 0$. Hence the point $(0, a)$ is a maximum point of the curve.

When $y'' = 0$, $2b^2 x^2 - 1 = 0$, yielding $x = \pm\sqrt{2}/2b$ as possible points of inflection. We have:



Hence the points $(\pm\sqrt{2}/2b, ae^{-1/2})$ are points of inflection.

22. The equilibrium constant K of a balanced chemical reaction changes with the absolute temperature T according to $K = K_0 e^{-\frac{1}{2} q(T-T_0)/T_0 T}$, where K_0 , q , and T_0 are constants. Find the percentage rate of change of K per degree of change of T .

The percentage rate of change of K per degree of change of T is given by $\frac{100}{K} \frac{dK}{dT} = 100 \frac{d}{dT} (\ln K)$. Then,

$$\ln K = \ln K_0 - \frac{1}{2} q \frac{T - T_0}{T_0 T} \quad \text{and} \quad 100 \frac{d}{dT} (\ln K) = -\frac{100q}{2T^2} = -\frac{50q}{T^2} \%$$

23. Discuss the damped-vibration curve $y = f(t) = e^{-\frac{1}{4}t} \sin 2\pi t$.

When $t = 0$, $y = 0$. The y intercept is thus 0.

When $y = 0$, we have $\sin 2\pi t = 0$ and $t = \dots, -\frac{3}{2}, -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$. These are the t intercepts.

When $t = \dots, -\frac{7}{4}, -\frac{5}{4}, -\frac{3}{4}, -\frac{1}{4}, \dots$, we have $\sin 2\pi t = 1$ and $y = e^{-\frac{1}{4}t}$. When $t = \dots, -\frac{5}{4}, -\frac{3}{4}, -\frac{1}{4}, \frac{1}{4}, \frac{3}{4}, \dots$, we have $\sin 2\pi t = -1$ and $y = -e^{-\frac{1}{4}t}$. The given curve oscillates between the two curves $y = e^{-\frac{1}{4}t}$ and $y = -e^{-\frac{1}{4}t}$, touching them at these points, as shown in Fig. 19-4.

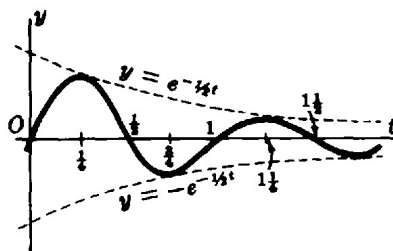


Fig. 19-4

Differentiation yields

$$y' = f'(t) = e^{-\frac{1}{4}t} (2\pi \cos 2\pi t - \frac{1}{4} \sin 2\pi t)$$

$$y'' = f''(t) = e^{-\frac{1}{4}t} [(\frac{1}{4} - 4\pi^2) \sin 2\pi t - 2\pi \cos 2\pi t]$$

When $y' = 0$, then $2\pi \cos 2\pi t - \frac{1}{4} \sin 2\pi t = 0$; that is, $\tan 2\pi t = 4\pi$. If $t = \xi = 0.237$ is the smallest positive angle satisfying this relation, then $t = \dots, \xi - \frac{1}{2}, \xi - 1, \xi - \frac{1}{2}, \xi, \xi + \frac{1}{2}, \xi + 1, \dots$ are the critical values.

For $n = 0, 1, 2, \dots$, $f''(\xi \pm \frac{1}{2}n)$ and $f''(\xi \pm \frac{n+1}{2})$ have opposite signs, whereas $f''(\xi \pm \frac{1}{2}n)$ and $f''(\xi \pm \frac{n+2}{2})$ have the same sign; hence, the critical values yield alternate maximum and minimum points of the curve. These points are slightly to the left of the points of contact with the curves $y = e^{-\frac{1}{4}t}$ and $y = -e^{-\frac{1}{4}t}$.

When $y'' = 0$, $\tan 2\pi t = \frac{2\pi}{\frac{1}{4} - 4\pi^2} = \frac{8\pi}{1 - 16\pi^2}$. If $t = \eta = 0.475$ is the smallest positive angle satisfying this relation, then $t = \dots, \eta - 1, \eta - \frac{1}{2}, \eta, \eta + \frac{1}{2}, \eta + 1, \dots$ are the possible points of inflection. These points, located slightly to the left of the points of intersection of the curve and the t axis, are points of inflection.

24. The equation $s = ce^{-bt} \sin(kt + \theta)$, where c , b , k , and θ are constants, represents damped vibratory motion. Show that $a = -2bv - (k^2 + b^2)s$, where $v = ds/dt$ and $a = dv/dt$.

$$v = \frac{ds}{dt} = ce^{-bt} [-b \sin(kt + \theta) + k \cos(kt + \theta)]$$

$$\begin{aligned} a = \frac{dv}{dt} &= ce^{-bt} [(b^2 - k^2) \sin(kt + \theta) - 2bk \cos(kt + \theta)] \\ &= ce^{-bt} \{-2b[-b \sin(kt + \theta) + k \cos(kt + \theta)] - (k^2 + b^2) \sin(kt + \theta)\} \\ &= -2bv - (k^2 + b^2)s \end{aligned}$$

Supplementary Problems

In Problems 25 to 35, find dy/dx .

25. $y = \ln(4x - 5)$

Ans. $4/(4x - 5)$

26. $y = \ln \sqrt{3 - x^2}$

Ans. $x/(x^2 - 3)$

27. $y = \ln 3x^5$ *Ans.* $5/x$
28. $y = \ln(x^2 + x - 1)^3$ *Ans.* $(6x + 3)/(x^2 + x - 1)$
29. $y = x \cdot \ln x - x$ *Ans.* $\ln x$
30. $y = \ln(\sec x + \tan x)$ *Ans.* $\sec x$
31. $y = \ln(\ln \tan x)$ *Ans.* $2/(\sin 2x \ln \tan x)$
32. $y = (\ln x^2)/x^2$ *Ans.* $(2 - 4 \ln x)/x^3$
33. $y = \frac{1}{3}x^5(\ln x - \frac{1}{3})$ *Ans.* $x^4 \ln x$
34. $y = x[\sin(\ln x) - \cos(\ln x)]$ *Ans.* $2 \sin(\ln x)$
35. $y = x \ln(4 + x^2) + 4 \arctan \frac{1}{2}x - 2x$ *Ans.* $\ln(4 + x^2)$
36. Find the equation of the line tangent to $y = \ln x$ at any one of its points (x_0, y_0) . Use the y intercept of the tangent line to obtain a simple construction for the tangent line.
Ans. $y - y_0 = (1/x_0)(x - x_0)$
37. Discuss and sketch: $y = x^2 \ln x$. *Ans.* minimum at $x = 1/\sqrt{e}$; inflection point at $x = 1/e^{3/2}$
38. Show that the angle of intersection of the curves $y = \ln(x - 2)$ and $y = x^2 - 4x + 3$ at the point $(3, 0)$ is $\phi = \arctan \frac{1}{3}$.

In Problems 39 to 46, find dy/dx .

39. $y = e^{5x}$ *Ans.* $5e^{5x}$
40. $y = e^{x^3}$ *Ans.* $3x^2 e^{x^3}$
41. $y = e^{\sin 3x}$ *Ans.* $3e^{\sin 3x} \cos 3x$
42. $y = 3^{-x^2}$ *Ans.* $-2x(3^{-x^2} \ln 3)$
43. $y = e^{-x} \cos x$ *Ans.* $-e^{-x}(\cos x + \sin x)$
44. $y = \arcsin e^x$ *Ans.* $e^x/\sqrt{1 - e^{2x}}$
45. $y = \tan^2 e^{3x}$ *Ans.* $6e^{3x} \tan e^{3x} \sec^2 e^{3x}$
46. $y = e^{e^x}$ *Ans.* $e^{(x + e^x)}$
47. If $y = x^2 e^x$, show that $y''' = (x^2 + 6x + 6)e^x$.
48. If $y = e^{-2x}(\sin 2x + \cos 2x)$, show that $y'' + 4y' + 8y = 0$.
49. Discuss and sketch: (a) $y = x^2 e^{-x}$ and (b) $y = x^2 e^{-x^2}$.
Ans. (a) maximum at $x = 2$; minimum at $x = 0$; inflection points at $x = 2 \pm \sqrt{2}$
 (b) maximum at $x = \pm 1$; minimum at $x = 0$; inflection points at $x = \pm 1.51$, $x = \pm 0.47$
50. Find the rectangle of maximum area, having one edge along the x axis, under the curve $y = e^{-x^2}$. (*Hint:* $A = 2xy = 2xe^{-x^2}$, where $P(x, y)$ is a vertex of the rectangle on the curve.) *Ans.* $A = \sqrt{2}/e$
51. Show that the curves $y = e^{ax}$ and $y = e^{ax} \cos ax$ are tangent at the points for which $x = 2n\pi/a$ ($n = 1, 2, 3, \dots$), and that the curves $y = e^{-ax/a^2}$ and $y = e^{ax} \cos ax$ are mutually perpendicular at the same points.

52. For the curve $y = xe^x$, show (a) $(-1, -1/e)$ is a relative minimum point, (b) $(-2, -2/e^2)$ is a point of inflection, and (c) the curve is concave downward to the left of the point of inflection, and concave upward to the right of it.

In Problems 53 to 56, use logarithmic differentiation to find dy/dx .

53. $y = x^x$ *Ans.* $x^x(1 + \ln x)$

54. $y = x^2 e^{2x} \cos 3x$ *Ans.* $x^2 e^{2x} \cos 3x(2/x + 2 - 3 \tan 3x)$

55. $y = x^{\ln x}$ *Ans.* $2x^{(\ln x - 1)} \ln x$

56. $y = x^{e^{-x^2}}$ *Ans.* $e^{-x^2} x^{e^{-x^2}} (1/x - 2x \ln x)$

57. Show (a) $\frac{d^n}{dx^n} (xe^x) = (x + n)e^x$; (b) $\frac{d^n}{dx^n} (x^{n-1} \ln x) = \frac{(n-1)!}{x}$.

Chapter 20

Differentiation of Hyperbolic Functions

DEFINITIONS OF HYPERBOLIC FUNCTIONS. For x any real number, except where noted, the hyperbolic functions are defined as

$$\begin{aligned}\sinh x &= \frac{e^x - e^{-x}}{2} & \coth x &= \frac{1}{\tanh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}, \quad x \neq 0 \\ \cosh x &= \frac{e^x + e^{-x}}{2} & \operatorname{sech} x &= \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}} \\ \tanh x &= \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} & \operatorname{csch} x &= \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}, \quad x \neq 0\end{aligned}$$

DIFFERENTIATION FORMULAS

$$\begin{aligned}31. \quad \frac{d}{dx} (\sinh x) &= \cosh x & 32. \quad \frac{d}{dx} (\cosh x) &= \sinh x \\ 33. \quad \frac{d}{dx} (\tanh x) &= \operatorname{sech}^2 x & 34. \quad \frac{d}{dx} (\coth x) &= -\operatorname{csch}^2 x \\ 35. \quad \frac{d}{dx} (\operatorname{sech} x) &= -\operatorname{sech} x \tanh x & 36. \quad \frac{d}{dx} (\operatorname{csch} x) &= -\operatorname{csch} x \coth x\end{aligned}$$

(See Problems 1 to 12.)

DEFINITIONS OF INVERSE HYPERBOLIC FUNCTIONS

$$\begin{aligned}\sinh^{-1} x &= \ln(x + \sqrt{1+x^2}) \quad \text{for all } x & \coth^{-1} x &= \frac{1}{2} \ln \frac{x+1}{x-1}, \quad x^2 > 1 \\ \cosh^{-1} x &= \ln(x + \sqrt{x^2-1}), \quad x \geq 1 & \operatorname{sech}^{-1} x &= \ln \frac{1 + \sqrt{1-x^2}}{x}, \quad 0 < x \leq 1 \\ \tanh^{-1} x &= \frac{1}{2} \ln \frac{1+x}{1-x}, \quad x^2 < 1 & \operatorname{csch}^{-1} x &= \ln \left(\frac{1}{x} + \frac{\sqrt{1+x^2}}{|x|} \right), \quad x \neq 0\end{aligned}$$

(Only principal values of $\cosh^{-1} x$ and $\operatorname{sech}^{-1} x$ are included here.)

DIFFERENTIATION FORMULAS

$$\begin{aligned}37. \quad \frac{d}{dx} (\sinh^{-1} x) &= \frac{1}{\sqrt{1+x^2}} & 38. \quad \frac{d}{dx} (\cosh^{-1} x) &= \frac{1}{\sqrt{x^2-1}}, \quad x > 1 \\ 39. \quad \frac{d}{dx} (\tanh^{-1} x) &= \frac{1}{1-x^2}, \quad x^2 < 1 & 40. \quad \frac{d}{dx} (\coth^{-1} x) &= \frac{1}{1-x^2}, \quad x^2 > 1 \\ 41. \quad \frac{d}{dx} (\operatorname{sech}^{-1} x) &= \frac{-1}{x\sqrt{1-x^2}}, \quad 0 < x < 1 & 42. \quad \frac{d}{dx} (\operatorname{csch}^{-1} x) &= \frac{-1}{|x|\sqrt{1+x^2}}, \quad x \neq 0\end{aligned}$$

(See Problems 13 to 19.)

Solved Problems

1. Prove that $\cosh^2 u - \sinh^2 u = 1$.

$$\cosh^2 u - \sinh^2 u = \left(\frac{e^u + e^{-u}}{2} \right)^2 - \left(\frac{e^u - e^{-u}}{2} \right)^2 = \frac{1}{4}(e^{2u} + 2 + e^{-2u}) - \frac{1}{4}(e^{2u} - 2 + e^{-2u}) = 1$$

2. Derive $\frac{d}{dx} (\sinh x) = \cosh x$.

$$\frac{d}{dx} (\sinh x) = \frac{d}{dx} \left(\frac{e^x - e^{-x}}{2} \right) = \frac{e^x + e^{-x}}{2} = \cosh x$$

In Problems 3 to 10, find dy/dx .

3. $y = \sinh 3x$: $\frac{dy}{dx} = \cosh 3x \frac{d}{dx} (3x) = 3 \cosh 3x$

4. $y = \cosh \frac{1}{2}x$: $\frac{dy}{dx} = \sinh \frac{1}{2}x \frac{d}{dx} \left(\frac{1}{2}x \right) = \frac{1}{2} \sinh \frac{1}{2}x$

5. $y = \tanh (1 + x^2)$: $\frac{dy}{dx} = \operatorname{sech}^2 (1 + x^2) \frac{d}{dx} (1 + x^2) = 2x \operatorname{sech}^2 (1 + x^2)$

6. $y = \coth \frac{1}{x}$: $\frac{dy}{dx} = -\operatorname{csch}^2 \frac{1}{x} \frac{d}{dx} \left(\frac{1}{x} \right) = \frac{1}{x^2} \operatorname{csch}^2 \frac{1}{x}$

7. $y = x \operatorname{sech} x^2$:

$$\begin{aligned} \frac{dy}{dx} &= x \frac{d}{dx} (\operatorname{sech} x^2) + \operatorname{sech} x^2 \frac{d}{dx} (x) = x(-\operatorname{sech} x^2 \tanh x^2)2x + \operatorname{sech} x^2 \\ &= -2x^2 \operatorname{sech} x^2 \tanh x^2 + \operatorname{sech} x^2 \end{aligned}$$

8. $y = \operatorname{csch}^2 (x^2 + 1)$:

$$\begin{aligned} \frac{dy}{dx} &= 2 \operatorname{csch} (x^2 + 1) \frac{d}{dx} [\operatorname{csch} (x^2 + 1)] = 2 \operatorname{csch} (x^2 + 1) [-\operatorname{csch} (x^2 + 1) \coth (x^2 + 1) \cdot 2x] \\ &= -4x \operatorname{csch}^2 (x^2 + 1) \coth (x^2 + 1) \end{aligned}$$

9. $y = \frac{1}{4} \sinh 2x - \frac{1}{2}x$: $\frac{dy}{dx} = \frac{1}{4}(\cosh 2x)2 - \frac{1}{2} = \frac{1}{2}(\cosh 2x - 1) = \sinh^2 x$

10. $y = \ln \tanh 2x$: $\frac{dy}{dx} = \frac{1}{\tanh 2x} (2 \operatorname{sech}^2 2x) = \frac{2}{\sinh 2x \cosh 2x} = 4 \operatorname{csch} 4x$

11. Find the coordinates of the minimum point of the catenary $y = a \cosh \frac{x}{a}$.

$$f'(x) = \frac{1}{a} \left(a \sinh \frac{x}{a} \right) = \sinh \frac{x}{a} \quad \text{and} \quad f''(x) = \frac{1}{a} \cosh \frac{x}{a} = \frac{1}{a} \frac{e^{x/a} + e^{-x/a}}{2}$$

When $f'(x) = \frac{e^{x/a} - e^{-x/a}}{2} = 0$, $x = 0$; and $f''(0) > 0$. Hence, the point $(0, a)$ is the minimum point.

12. Examine (a) $y = \sinh x$, (b) $y = \cosh x$, and (c) $y = \tanh x$ for points of inflection.

(a) $f'(x) = \cosh x$, $f''(x) = \sinh x$, and $f'''(x) = \cosh x$.

$f'''(x) = \sinh x = 0$ when $x = 0$, and $f'''(0) \neq 0$. Hence, the point $(0, 0)$ is a point of inflection.

(b) $f'(x) = \sinh x$, and $f''(x) = \cosh x \neq 0$ for all values of x . There is no point of inflection.

(c) $f'(x) = \operatorname{sech}^2 x$, $f''(x) = -2 \operatorname{sech}^2 x \tanh x = -2 \frac{\sinh x}{\cosh^3 x}$, and $f'''(x) = \frac{4 \sinh^2 x - 2}{\cosh^4 x}$.
 $f''(x) = 0$ when $x = 0$, and $f'''(0) \neq 0$. The point $(0, 0)$ is a point of inflection.

13. Derive: (a) $\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$, for all x

$$(b) \operatorname{sech}^{-1} x = \cosh^{-1} \frac{1}{x} = \ln \frac{1 + \sqrt{1 - x^2}}{x}, \text{ for } 0 < x \leq 1$$

(a) Let $\sinh^{-1} x = y$; then $x = \sinh y = \frac{1}{2}(e^y - e^{-y})$ or, after multiplication by $2e^y$, $e^{2y} - 2xe^y - 1 = 0$. Solving for e^y yields $e^y = x + \sqrt{x^2 + 1}$, since $e^y > 0$. Thus, $y = \ln(x + \sqrt{x^2 + 1})$.

(b) Let $\operatorname{sech}^{-1} x = y$; then $x = \operatorname{sech} y = \frac{1}{\cosh y}$, so $\cosh y = \frac{1}{x}$. Hence $y = \cosh^{-1} \frac{1}{x} = \operatorname{sech}^{-1} x$. Also, $x = \operatorname{sech} y = \frac{2}{e^y + e^{-y}}$, from which $e^{2y}x - 2e^y + x = 0$. Solving for e^y yields $e^y = \frac{1 + \sqrt{1 - x^2}}{x}$ for $y \geq 0$. Thus, $y = \ln \frac{1 + \sqrt{1 - x^2}}{x}$, $0 < x \leq 1$.

14. Derive $\frac{d}{dx} (\sinh^{-1} x) = \frac{1}{\sqrt{1 + x^2}}$.

Let $y = \sinh^{-1} x$. Then $\sinh y = x$ and differentiation yields $\cosh y \frac{dy}{dx} = 1$; so

$$\frac{dy}{dx} = \frac{1}{\cosh y} = \frac{1}{\sqrt{1 + \sinh^2 y}} = \frac{1}{\sqrt{1 + x^2}}$$

In Problems 15 to 19, find dy/dx .

15. $y = \sinh^{-1} 3x$: $\frac{dy}{dx} = \frac{1}{\sqrt{(3x)^2 + 1}} \frac{d}{dx} (3x) = \frac{3}{\sqrt{9x^2 + 1}}$

16. $y = \cosh^{-1} e^x$: $\frac{dy}{dx} = \frac{1}{\sqrt{e^{2x} - 1}} \frac{d}{dx} (e^x) = \frac{e^x}{\sqrt{e^{2x} - 1}}$

17. $y = 2 \tanh^{-1} (\tan \frac{1}{2} x)$: $\frac{dy}{dx} = 2 \frac{1}{1 - \tan^2 \frac{1}{2} x} \frac{d}{dx} (\tan \frac{1}{2} x)$
 $= 2 \frac{1}{1 - \tan^2 \frac{1}{2} x} (\sec^2 \frac{1}{2} x) (\frac{1}{2}) = \frac{\sec^2 \frac{1}{2} x}{1 - \tan^2 \frac{1}{2} x} = \sec x$

18. $y = \coth^{-1} \frac{1}{x}$: $\frac{dy}{dx} = \frac{1}{1 - (1/x)^2} \frac{d}{dx} \left(\frac{1}{x} \right) = \frac{-1/x^2}{1 - 1/x^2} = \frac{-1}{x^2 - 1}$

19. $y = \operatorname{sech}^{-1} (\cos x)$: $\frac{dy}{dx} = \frac{-1}{\cos x \sqrt{1 - \cos^2 x}} \frac{d}{dx} (\cos x) = \frac{\sin x}{\cos x \sqrt{1 - \cos^2 x}} = \sec x$

Supplementary Problems

20. (a) Sketch the curves of $y = e^x$ and $y = -e^{-x}$, and average the ordinates of the two curves for various values of x to obtain points on $y = \sinh x$. Complete the curve.
 (b) Proceed as in (a), using $y = e^x$ and $y = e^{-x}$ to obtain the graph of $y = \cosh x$.

21. For the hyperbola $x^2 - y^2 = 1$ in Fig. 20-1, show that (a) $P(\cosh u, \sinh u)$ is a point on the hyperbola; (b) the tangent line at A intersects the line OP at $T(1, \tanh u)$.

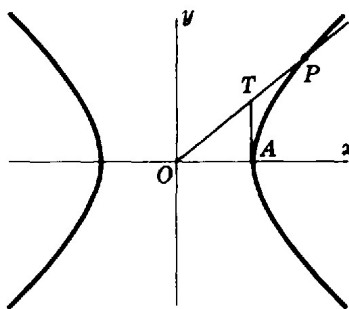


Fig. 20-1

22. Show: (a) $\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y$
 (b) $\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y$
 (c) $\sinh 2x = 2 \sinh x \cosh x$
 (d) $\cosh 2x = \cosh^2 x + \sinh^2 x = 2 \cosh^2 x - 1 = 2 \sinh^2 x + 1$
 (e) $\tanh 2x = \frac{2 \tanh x}{1 + \tanh^2 x}$

In Problems 23 to 28, find dy/dx .

23. $y = \sinh \frac{1}{4}x$ *Ans.* $\frac{1}{4} \cosh \frac{1}{4}x$ 24. $y = \cosh^2 3x$ *Ans.* $3 \sinh 6x$
 25. $y = \tanh 2x$ *Ans.* $2 \operatorname{sech}^2 2x$ 26. $y = \ln \cosh x$ *Ans.* $\tanh x$
 27. $y = \operatorname{arc tan} \sinh x$ *Ans.* $\operatorname{sech} x$ 28. $y = \ln \sqrt{\tanh 2x}$ *Ans.* $2 \operatorname{csch} 4x$
29. Show: (a) If $y = a \cosh \frac{x}{a}$, then $y'' = \frac{1}{a} \sqrt{1 + (y')^2}$.
 (b) If $y = A \cosh bx + B \sinh bx$, where b , A , and B are constants, then $y'' = b^2 y$.
30. Show: (a) $\cosh^{-1} u = \ln(u + \sqrt{u^2 - 1})$, $u \geq 1$, and (b) $\tanh^{-1} u = \frac{1}{2} \ln \frac{1+u}{1-u}$, $u^2 < 1$.
31. (a) Trace the curve $y = \sinh^{-1} x$ by reflecting the curve $y = \sinh x$ in the 45° line.
 (b) Trace the principal branch of $y = \cosh^{-1} x$ by reflecting the right half of $y = \cosh x$ in the 45° line.
32. Derive differentiation formulas 32 to 36, 38 to 40, and 42.

In Problems 33 to 36, find dy/dx .

33. $y = \sinh^{-1} \frac{1}{2}x$ *Ans.* $\frac{1}{\sqrt{x^2 + 4}}$ 34. $y = \cosh^{-1} \frac{1}{x}$ *Ans.* $-\frac{1}{x\sqrt{1-x^2}}$
 35. $y = \tanh^{-1}(\sin x)$ *Ans.* $\sec x$
 36. $x = a \operatorname{sech}^{-1} \frac{y}{a} - \sqrt{a^2 - y^2}$ *Ans.* $-\frac{y}{\sqrt{a^2 - y^2}}$